



### 4.1 角的定义及弧度(整合习题)

### (4.1 Definition and Radian Measure of Angles (Integrated Exercises))

### 一、选择题(Multiple Choice)

1.  $1010^\circ$  是第 ( ) 象限角。

( $1010^\circ$  is an angle in the ( ) quadrant.)

- A. 一(I) B. 二(II) C. 三(III) D. 四(IV)

2. 下列角中终边与  $320^\circ$  相同的角是 ( )。

(Which of the following angles has the same terminal side as  $320^\circ$ ?)

- A.  $40^\circ$  B.  $-40^\circ$  C.  $620^\circ$  D.  $-620^\circ$

3.  $-1060^\circ$  写成  $\alpha + 2k\pi$  ( $k \in \mathbb{Z}, 0 \leq \alpha < 2\pi$ ) 的形式是 ( )。

(Express  $-1060^\circ$  in the form  $\alpha + 2k\pi$  ( $k \in \mathbb{Z}, 0 \leq \alpha < 2\pi$ ) ( ).)

- A.  $-\frac{\pi}{9} - 6\pi$  B.  $\frac{2\pi}{9} - 6\pi$  C.  $\frac{\pi}{9} - 6\pi$  D.  $-\frac{2\pi}{9} - 6\pi$

## 二、填空题(Fill in the Blanks)

1.  $45^\circ =$              $\pi$ ;  $315^\circ =$              $\pi$ ;  $-20^\circ =$              $\pi$ ;

2.  $\frac{13}{3}\pi = \underline{\hspace{2cm}}^\circ$ ;  $\frac{3}{5}\pi = \underline{\hspace{2cm}}^\circ$ .

3. 把下列角写成  $360^\circ \cdot k + \alpha (k \in \mathbb{Z}, 0^\circ \leq \alpha < 360^\circ)$  或  $2k\pi + \alpha (k \in \mathbb{Z}, 0 \leq \alpha < 2\pi)$  的形式:

(Express the following angles in the form  $360^\circ \cdot k + \alpha$  ( $k \in \mathbb{Z}$ ,  $0^\circ \leq \alpha < 360^\circ$ ) or  $2k\pi + \alpha$  ( $k \in \mathbb{Z}$ ,  $0 \leq \alpha < 2\pi$ ):)

380° ;

$$\frac{11}{4}\pi = \underline{\hspace{2cm}}; \underline{\hspace{2cm}}; \circ$$

4.  $-1130^\circ$  在第\_\_\_\_\_象限; $1495^\circ$  在第\_\_\_\_\_象限;

$-\frac{19}{4}\pi$  在第\_\_\_\_象限。

( $-1130^\circ$  lies in the \_\_\_\_ quadrant;  $1495^\circ$  lies in the \_\_\_\_ quadrant;  $-\frac{19}{4}\pi$  lies in the \_\_\_\_ quadrant.)

5. 锐角是第\_\_\_\_象限角;钝角是第\_\_\_\_象限角。

(An acute angle is a \_\_ quadrant angle; an obtuse angle is a \_\_ quadrant angle.)

### 三、计算题(Calculation)

1. 计算:  $220^\circ = \_\_\pi$ ;  $20^\circ = \_\_\pi$ 。

2. 把下列角写成  $2k\pi + \alpha (k \in \mathbb{Z}, 0 \leq \alpha < 2\pi)$  或  $k \cdot 360^\circ + \alpha (k \in \mathbb{Z}, 0^\circ \leq \alpha < 360^\circ)$  的形式: (Express the following angles in the form  $2k\pi + \alpha (k \in \mathbb{Z}, 0 \leq \alpha < 2\pi)$  or  $k \cdot 360^\circ + \alpha (k \in \mathbb{Z}, 0^\circ \leq \alpha < 360^\circ)$ .)

(1)  $\frac{11\pi}{4}$

(2)  $1160^\circ$

3. 判断下列角的终边在第几象限: (Determine which quadrant the terminal side of each of the following angles lies in:)

(1)  $-70^\circ$  \_\_\_\_\_ (2)  $595^\circ$  \_\_\_\_\_ (3)  $-960^\circ$  \_\_\_\_\_



## 4.2 三角函数的定义及关系(整合习题)

### (4.2 Definition and Relations of Trigonometric Functions (Integrated Exercises))

#### 一、选择题(Multiple Choice)

1. 角  $\alpha$  的终边经过点  $P(-4,3)$ , 则  $\cos\alpha = ( )$ 。

(The terminal side of angle  $\alpha$  passes through point  $P(-4,3)$ , then  $\cos\alpha = ( )$ .)

- A.  $-\frac{4}{3}$  B.  $\frac{4}{5}$  C.  $-\frac{4}{5}$  D.  $\frac{3}{5}$

2. 如果  $6\sin\alpha \cdot \cos\alpha < 0$ , 那么  $\alpha$  在  $( )$ 。

(If  $6\sin\alpha \cdot \cos\alpha < 0$ , then  $\alpha$  lies in  $( )$ .)

- A. 第一、二象限(Quadrants I and II) B. 第一、三象限(Quadrants I and III)  
C. 第一、四象限(Quadrants I and IV) D. 第二、四象限(Quadrants II and IV)

3. 下列运算正确的是  $( )$ 。

(Which of the following operations is correct  $( )$ ?)

- A.  $\cos\frac{\pi}{3} = \frac{1}{2}$  B.  $\sin\frac{\pi}{4} = \frac{\sqrt{3}}{2}$  C.  $\tan\frac{\pi}{3} = \frac{\sqrt{2}}{2}$  D.  $\sin\frac{\pi}{2} = 0$

4. 已知  $\cos\alpha = -\frac{1}{4}$ ,  $\alpha$  在第二象限, 则  $\sin\alpha = ( )$ 。

(Given  $\cos\alpha = -\frac{1}{4}$  and  $\alpha$  is in Quadrant II, then  $\sin\alpha = ( )$ .)

- A.  $\frac{\sqrt{15}}{4}$  B.  $-\frac{\sqrt{15}}{4}$  C.  $\frac{1}{4}$  D.  $-\frac{1}{4}$

5. 已知  $\tan\alpha = -\frac{3}{4}$ ,  $\alpha$  在第二象限, 则  $\cos\alpha = ( )$ 。

(Given  $\tan\alpha = -\frac{3}{4}$  and  $\alpha$  is in Quadrant II, then  $\cos\alpha = ( )$ .)

- A.  $\frac{4}{5}$  B.  $-\frac{4}{5}$  C.  $\frac{3}{5}$  D.  $-\frac{3}{5}$

6. 已知  $\sin\alpha = \frac{4}{5}$ ,  $\cos\alpha = \frac{3}{5}$ , 那么  $\tan\alpha = ( )$ 。

(Given  $\sin\alpha = \frac{4}{5}$  and  $\cos\alpha = \frac{3}{5}$ , then  $\tan\alpha = ( )$ .)

A.  $\frac{4}{3}$  B.  $\frac{3}{4}$  C.  $\frac{4}{5}$  D.  $\frac{3}{5}$

7. 已知  $\sin\alpha = -\frac{3}{5}$ ,  $\alpha$  在第三象限, 则  $\cos\alpha = ( )$ 。

(Given  $\sin\alpha = -\frac{3}{5}$  and  $\alpha$  is in Quadrant III, then  $\cos\alpha = ( )$ .)

A.  $-\frac{3}{5}$  B.  $\frac{4}{5}$  C.  $-\frac{4}{5}$  D.  $\frac{4}{3}$

## 二、填空题(Fill in the Blanks)

1. 判断下列三角函数值的正负:

(Determine the sign (positive/negative) of the following trigonometric values:)

(1)  $\cos(-440^\circ)$  \_\_\_\_\_; (2)  $\tan\left(-\frac{19\pi}{8}\right)$  \_\_\_\_\_; (3)  $\sin 158^\circ$  \_\_\_\_\_;

(4)  $\sin 105^\circ$  \_\_\_\_\_; (5)  $\cot 210^\circ$  \_\_\_\_\_; (6)  $\cos\left(-\frac{1}{8}\pi\right)$  \_\_\_\_\_。

2. 计算:  $\sin(-80^\circ) + \cos 10^\circ + \tan 50^\circ =$  \_\_\_\_\_。

3. 计算:  $\cos\frac{\pi}{3} + 3\tan\frac{\pi}{6} - \sin\frac{\pi}{6} =$  \_\_\_\_\_。

4. 已知  $\tan\alpha = -\sqrt{2}$ ,  $\alpha$  在第二象限, 则  $\cos\alpha =$  \_\_\_\_\_。

\*(Given \_\_\_\_\_ and \_\_\_\_\_ is in Quadrant II, then .)

5. 已知  $\tan\alpha = 3$ , 则  $\frac{3\sin\alpha - 2\cos\alpha}{2\sin\alpha + 3\cos\alpha} =$  \_\_\_\_\_。

(Given \_\_\_\_\_, then .)

## 三、计算题(Calculation)

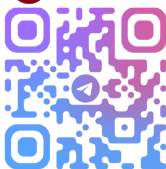
1. 判断下列三角函数值的正负:

(Determine the sign (positive/negative) of the following trigonometric values:)

(1)  $\sin 220^\circ$  \_\_\_\_\_ (2)  $\cos 98^\circ$  \_\_\_\_\_

(3)  $\tan\frac{4\pi}{5}$  \_\_\_\_\_ (4)  $\cot\frac{23\pi}{6}$  \_\_\_\_\_

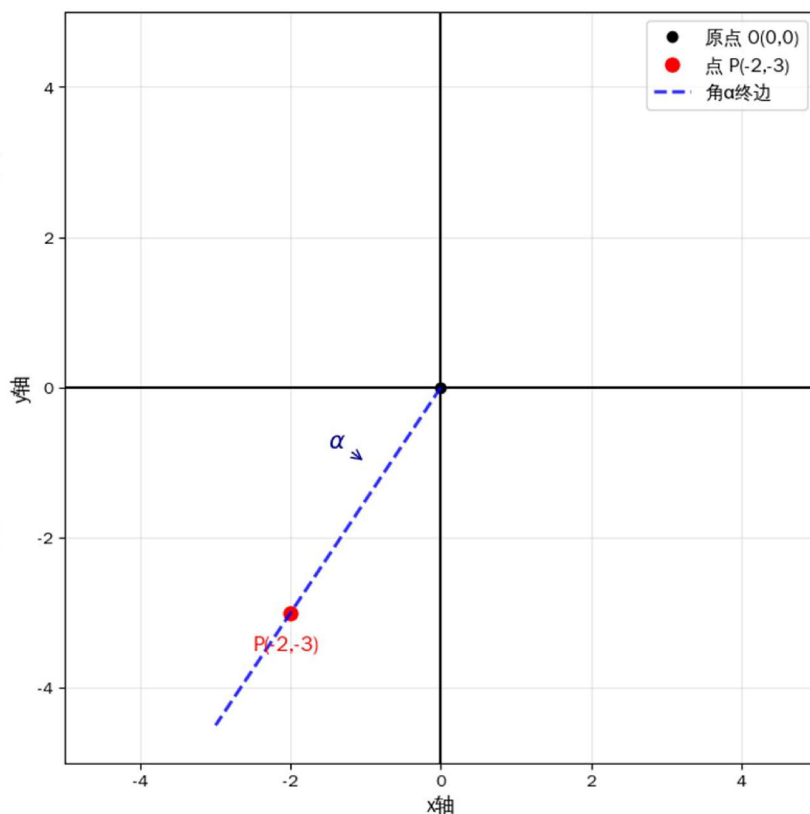




2. 已知角  $\alpha$  的终边经过点  $P(-2, -3)$ ，求角  $\alpha$  的六个三角函数值。

(The terminal side of angle  $\alpha$  passes through point  $P(-2, -3)$ , find the six trigonometric values of  $\alpha$ .)

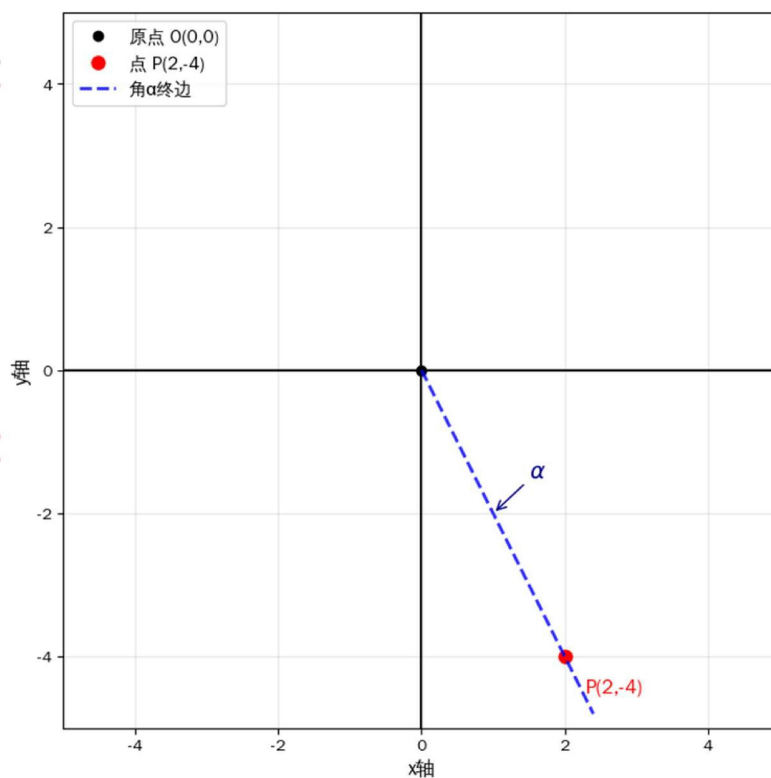
点  $P(-2, -3)$  及角  $\alpha$  终边示意图



3. 已知角  $\alpha$  的终边经过点  $P(2, -4)$ , 求角  $\alpha$  的六个三角函数值。

(The terminal side of angle  $\alpha$  passes through point  $P(2, -4)$ , find the six trigonometric values of  $\alpha$ .)

点  $P(2, -4)$  及角  $\alpha$  终边示意图

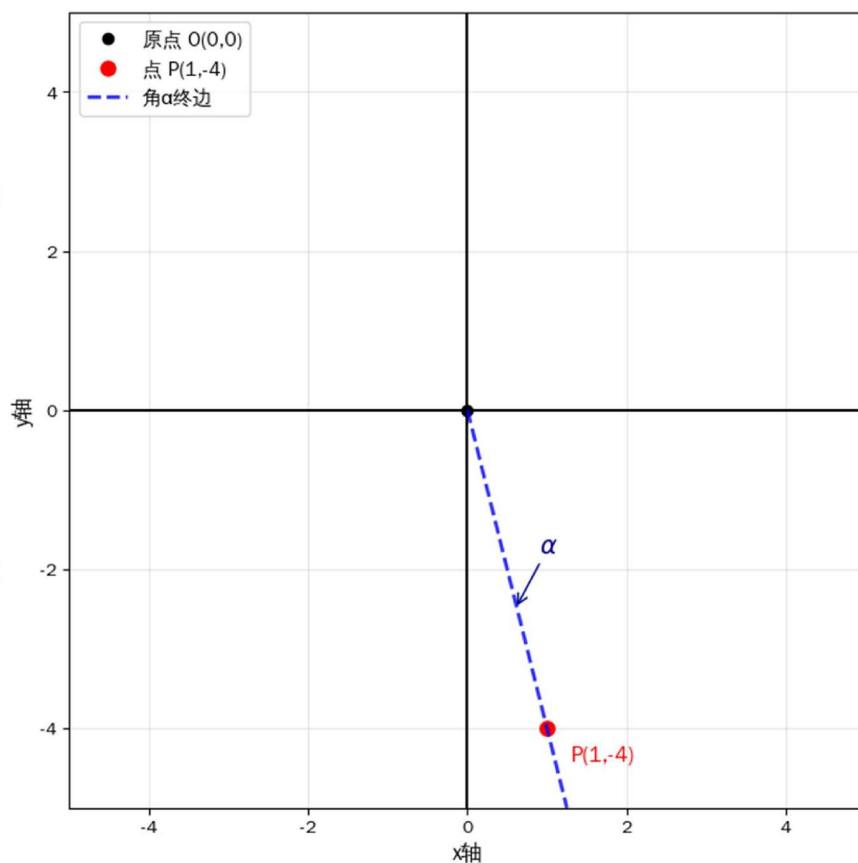




4. 已知角  $\alpha$  的终边经过点  $P(1, -4)$ , 求角  $\alpha$  的六个三角函数值。

(The terminal side of angle  $\alpha$  passes through point  $P(1, -4)$ , find the six trigonometric values of  $\alpha$ .)

点  $P(1, -4)$  及角  $\alpha$  终边示意图



5. 已知  $\sin\alpha = -\frac{3}{5}$ ,  $\alpha$  是第三象限角, 求  $\cos\alpha$ 、 $\tan\alpha$ 、 $\cot\alpha$  的值。

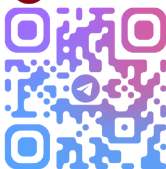
(Given  $\sin\alpha = -\frac{3}{5}$  and  $\alpha$  is a Quadrant III angle, find the values of  $\cos\alpha$ ,  $\tan\alpha$ , and  $\cot\alpha$ .)

6. 已知  $\sin\alpha = \frac{\sqrt{3}}{2}$ , 角  $\alpha$  在第二象限, 求  $\cos\alpha$ 、 $\tan\alpha$ 、 $\cot\alpha$  的值。

(Given  $\sin\alpha = \frac{\sqrt{3}}{2}$  and  $\alpha$  is in Quadrant II, find the values of  $\cos\alpha$ ,  $\tan\alpha$ , and  $\cot\alpha$ .)

7. 已知  $\tan\alpha = \frac{4}{5}$ ,  $\alpha \in \left(\pi, \frac{3}{2}\pi\right)$ , 求  $\cos\alpha$ 、 $\sin\alpha$  的值。

(Given  $\tan\alpha = \frac{4}{5}$  and  $\alpha \in \left(\pi, \frac{3}{2}\pi\right)$ , find the values of  $\cos\alpha$  and  $\sin\alpha$ .)



### 4.3 三角函数的诱导公式(整合习题)

#### (4.3 Induction Formulas of Trigonometric Functions (Integrated Exercises))

#### 一、选择题(Multiple Choice)

1.  $\sin \frac{5\pi}{6} = ( )$ 。

A.  $-\frac{1}{2}$  B.  $\frac{1}{2}$  C.  $\frac{\sqrt{3}}{2}$  D.  $-\frac{\sqrt{3}}{2}$

2.  $\tan\left(-\frac{11\pi}{6}\right) = ( )$ 。

A.  $-\sqrt{3}$  B.  $-\frac{\sqrt{3}}{3}$  C.  $\frac{\sqrt{3}}{3}$  D.  $\sqrt{3}$

3.  $\sin \alpha = ( )$ 。

A.  $\sin(\pi - \alpha)$  B.  $\sin(\pi + \alpha)$  C.  $\sin(-\alpha)$  D.  $\sin\left(\frac{\pi}{2} + \alpha\right)$

#### 二、填空题(Fill in the Blanks)

1. 求下列三角函数值：(Find the values of the following trigonometric functions:)

$\tan 600^\circ = \underline{\hspace{2cm}}; \sin\left(-\frac{17\pi}{4}\right) = \underline{\hspace{2cm}}; \cos(-945^\circ) = \underline{\hspace{2cm}};$

$\cos 210^\circ = \underline{\hspace{2cm}}; \tan 150^\circ = \underline{\hspace{2cm}}; \sin\left(-\frac{26\pi}{3}\right) = \underline{\hspace{2cm}};$

$\cot 780^\circ = \underline{\hspace{2cm}}; \tan \frac{19\pi}{4} = \underline{\hspace{2cm}}; \cos \frac{10\pi}{3} = \underline{\hspace{2cm}}。$

2. 如果  $\sin \alpha = \frac{1}{2}$ ,  $\alpha$  在第二象限, 那么  $\alpha = \underline{\hspace{2cm}}。$

(If  $\sin \alpha = \frac{1}{2}$  and  $\alpha$  is in Quadrant II, then.)

#### 三、计算题(Calculation)

1. 求下列三角函数值：

(Find the values of the following trigonometric functions:)

(1)  $\sin 235^\circ$  \_\_\_\_\_ (2)  $\cos \frac{13\pi}{3}$  \_\_\_\_\_ (3)  $\cos 400^\circ$  \_\_\_\_\_

(4)  $\sin \frac{11\pi}{4}$  \_\_\_\_\_ (5)  $\tan\left(-\frac{13\pi}{6}\right)$  \_\_\_\_\_



2. 化简求值:  $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(-\alpha - 2\pi)$ 。

(Simplify and evaluate:  $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(-\alpha - 2\pi)$ .)

3. 化简求值:  $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(\pi - \alpha)$ 。

(Simplify and evaluate:  $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(\pi - \alpha)$ .)



## 4.4 正弦函数和余弦函数的图像和性质(整合习题)

(4.4 Graphs and Properties of Sine and Cosine Functions (Integrated Exercises))

### 一、选择题(Multiple Choice)

1. 下列是  $y = \sin x$  的增区间的是 ( )。

(Which of the following is an increasing interval of  $y = \sin x$  ( )?)

A.  $[-\frac{\pi}{2}, \frac{\pi}{2}]$  B.  $[\frac{\pi}{2}, \frac{3\pi}{2}]$  C.  $[0, \pi]$  D.  $[\pi, 2\pi]$

2.  $y = 3\cos(\frac{1}{4}x)$  ( $x \in \mathbb{R}$ ) 是 ( )。

( $y = 3\cos(\frac{1}{4}x)$  ( $x \in \mathbb{R}$ ) is a ( ) function.)

A. 奇函数(Odd) B. 偶函数(Even)

C. 既是奇函数又是偶函数(Both odd and even)

D. 非奇非偶函数(Neither odd nor even)

3.  $y = 3\sin(\frac{1}{4}x + \frac{\pi}{6})$  的最小正周期是 ( )。

(The minimum positive period of  $y = 3\sin(\frac{1}{4}x + \frac{\pi}{6})$  is ( ).)

A.  $2\pi$  B.  $4\pi$  C.  $6\pi$  D.  $8\pi$

4.  $y = 4\cos 3x$  ( $x \in \mathbb{R}$ ) 的最小正周期是 ( )。

(The minimum positive period of  $y = 4\cos 3x$  ( $x \in \mathbb{R}$ ) is ( ).)

A.  $2\pi$  B.  $\frac{2\pi}{3}$  C.  $4\pi$  D.  $\frac{\pi}{3}$

5. 函数  $y = -4\sin x$  在  $[0, 2\pi]$  上的单调性是 ( )。

(The monotonicity of the function  $y = -4\sin x$  on  $[0, 2\pi]$  is ( ).)

A. 在  $[0, \pi]$  上是增函数, 在  $[\pi, 2\pi]$  上是减函数

(Increasing on  $[0, \pi]$  and decreasing on  $[\pi, 2\pi]$ )

B. 在  $[\frac{\pi}{2}, \frac{3\pi}{2}]$  上是增函数, 在  $[0, \frac{\pi}{2}] \cup [\frac{3\pi}{2}, 2\pi]$  上是减函数

(Increasing on  $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$  and decreasing on  $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$ )

C. 在  $\left[0, \frac{\pi}{2}\right]$  上是增函数, 在  $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$  上是减函数

(Increasing on  $\left[0, \frac{\pi}{2}\right]$  and decreasing on  $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ )

D. 在  $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$  上是增函数, 在  $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$  上是减函数

(Increasing on  $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$  and decreasing on  $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ )

## 二、填空题(Fill in the Blanks)

1.  $y = \sin x$  的最小正周期是\_\_\_\_;最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = \sin x$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

2.  $y = 4\sin x$  的最小正周期是\_\_\_\_;最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = 4\sin x$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

3.  $y = 4\sin \frac{1}{3}x$  的最小正周期是\_\_\_\_;最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = 4\sin \frac{1}{3}x$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

4.  $y = 4\sin\left(\frac{1}{3}x + \frac{\pi}{7}\right)$  的最小正周期是\_\_\_\_;

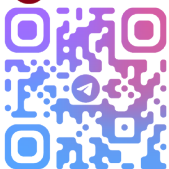
最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = 4\sin\left(\frac{1}{3}x + \frac{\pi}{7}\right)$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

5.  $y = 4\sin \frac{1}{3}x + 2$  的最小正周期是\_\_\_\_;

最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = 4\sin \frac{1}{3}x + 2$  is \_\_\_\_; its maximum and



minimum values are \_\_\_\_.)

6.  $y = -3\cos x$  的最小正周期是\_\_\_\_;最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = -3\cos x$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

7.  $y = -3\cos 4x$  的最小正周期是\_\_\_\_;最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = -3\cos 4x$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

8.  $y = -3\cos\left(4x - \frac{\pi}{7}\right)$  的最小正周期是\_\_\_\_;

最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = -3\cos\left(4x - \frac{\pi}{7}\right)$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

9.  $y = -3\cos 4x + 4$  的最小正周期是\_\_\_\_;

最大值和最小值是\_\_\_\_,\_\_\_\_。

(The minimum positive period of  $y = -3\cos 4x + 4$  is \_\_\_\_; its maximum and minimum values are \_\_\_\_.)

10.  $y = 3\cos x$  的奇偶性是\_\_\_\_;在  $(-\pi, \pi)$  上的减区间是\_\_\_\_。

(The parity of  $y = 3\cos x$  is \_\_\_\_; its decreasing interval on  $(-\pi, \pi)$  is \_\_\_\_.)

11.  $y = \sin x + 2$  的奇偶性是\_\_\_\_;在  $(0, \pi)$  上的增区间是\_\_\_\_。

(The parity of  $y = \sin x + 2$  is \_\_\_\_; its increasing interval on  $(0, \pi)$  is \_\_\_\_.)

12.  $y = 5\cos x - 4$  的奇偶性是\_\_\_\_;

$y = 5\sin x - 4$  的奇偶性是\_\_\_\_。

(The parity of  $y = 5\cos x - 4$  is \_\_\_, the parity of  $y = 5\sin x - 4$  is \_\_\_\_.)

13.  $y = \cos\left(x - \frac{\pi}{8}\right)$  的奇偶性是\_\_\_\_\_;

$y = \sin\left(x - \frac{\pi}{8}\right)$  的奇偶性是\_\_\_\_\_。

(The parity of  $y = \cos\left(x - \frac{\pi}{8}\right)$  is \_; the parity of  $y = \sin\left(x - \frac{\pi}{8}\right)$  is \_.)

### 三、计算题(Calculation)

1. 求下列函数的最小正周期:

(Find the minimum positive period of the following functions:)

(1)  $y = 6\sin x \ (x \in \mathbb{R})$

(2)  $y = \cos 4x \ (x \in \mathbb{R})$

(3)  $y = \sin\left(\frac{3}{4}x + \frac{\pi}{5}\right) \ (x \in \mathbb{R})$



2. 写出下列函数的最大值和最小值:

(Write the maximum and minimum values of the following functions:)

(1)  $y = \cos 3x$

(2)  $y = \sin 3x + 2$

(3)  $y = 4\sin x$

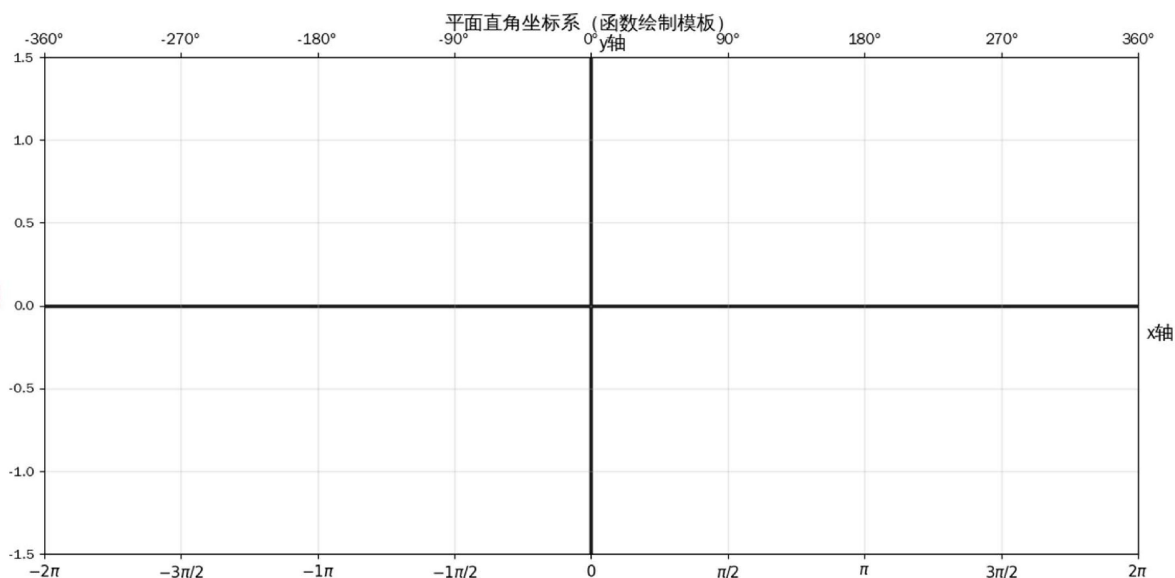


#### 四、画图题(Graphing)

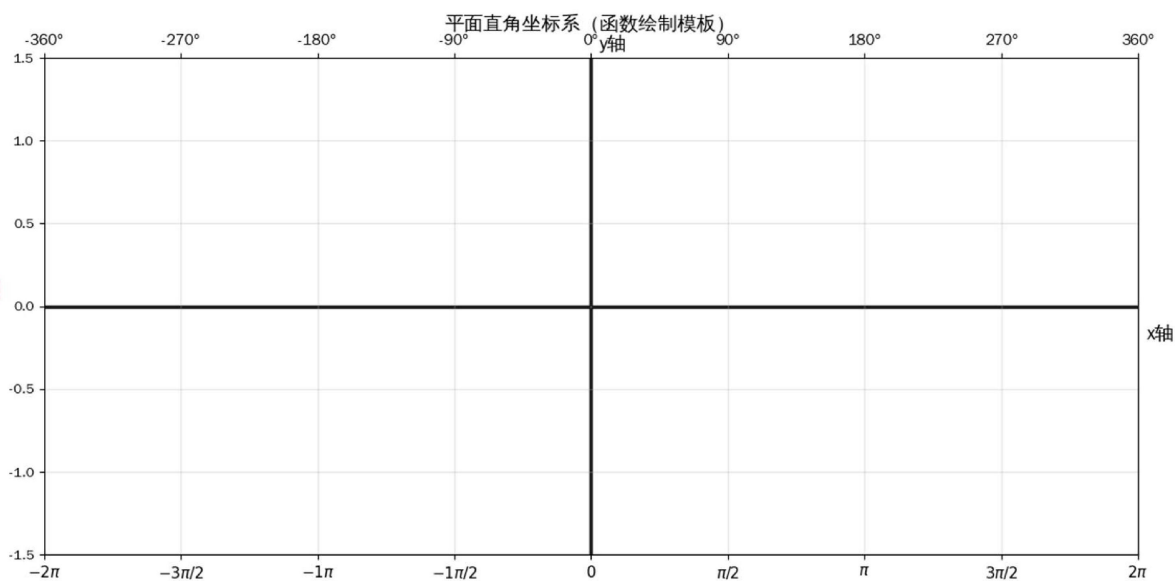
画出下列函数的图像：

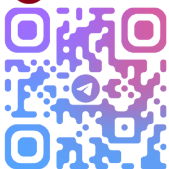
(Sketch the graphs of the following functions:)

(1)  $y = -2\sin x$ ;



(2)  $y = 4\cos x$





## 4.5 正切函数和余切函数的图像和性质(整合习题)

(4.5 Graphs and Properties of Tangent and Cotangent Functions (Integrated Exercises))

### 一、选择题(Multiple Choice)

1.  $y = 4\tan \frac{x}{5}$  的最小正周期是 ( ).

(The minimum positive period of  $y = 4\tan \frac{x}{5}$  is ( ).)

A.  $2\pi$  B.  $3\pi$  C.  $4\pi$  D.  $5\pi$

2.  $y = \tan\left(x + \frac{\pi}{5}\right)$  的定义域是 ( ).

(The domain of  $y = \tan\left(x + \frac{\pi}{5}\right)$  is ( ).)

A.  $\mathbb{R}$  B.  $\{x \mid x \neq k\pi + \frac{\pi}{2}, k \in \mathbb{Z}\}$  C.  $\{x \mid x \neq k\pi + \frac{3\pi}{10}, k \in \mathbb{Z}\}$  D.  $\{x \mid x \neq \frac{\pi}{5}\}$

3.  $y = \tan\left(3x + \frac{\pi}{4}\right)$  的奇偶性是 ( ).

(The parity of  $y = \tan\left(3x + \frac{\pi}{4}\right)$  is ( ).)

A. 奇函数(Odd) B. 偶函数(Even) C. 非奇非偶函数(Neither odd nor even) D.

无法确定(Cannot be determined)

4.  $y = \cot 4x$  的奇偶性是 ( ).

(The parity of  $y = \cot 4x$  is ( ).)

A. 奇函数(Odd) B. 偶函数(Even) C. 非奇非偶函数(Neither odd nor even) D.

无法确定(Cannot be determined)

### 二、填空题(Fill in the Blanks)

1.  $y = \tan 3x$  的最小正周期是\_\_\_\_\_;

定义域是\_\_\_\_\_。

(The minimum positive period of  $y = \tan 3x$  is \_\_\_\_\_; its domain is \_\_\_\_\_.)

2.  $y = -5\tan \frac{x}{3}$  的最小正周期是\_\_\_\_\_;

定义域是\_\_\_\_\_。

(The minimum positive period of  $y = -6\tan\frac{x}{4}$  is \_\_\_\_; its domain is \_\_\_\_.)

3.  $y = \frac{1}{3}\tan\left(x + \frac{\pi}{7}\right)$  的最小正周期是\_\_\_\_;

定义域是\_\_\_\_\_。

(The minimum positive period of  $y = \frac{1}{3}\tan\left(x + \frac{\pi}{7}\right)$  is \_\_\_\_; its domain is \_\_\_\_.)

4.  $y = \cot\frac{x}{3}$  的最小正周期是\_\_\_\_;

定义域是\_\_\_\_\_。

(The minimum positive period of  $y = \cot\frac{x}{3}$  is \_\_\_\_; its domain is \_\_\_\_.)

5.  $y = \cot 3x + 2$  的最小正周期是\_\_\_\_;

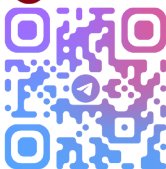
定义域是\_\_\_\_\_。

(The minimum positive period of  $y = \cot 3x + 2$  is \_\_\_\_; its domain is \_\_\_\_.)

### 三、计算题(Calculation)

求函数  $y = \tan\left(\frac{1}{3}x + \frac{\pi}{4}\right)$  的定义域和最小正周期。

(Find the domain and minimum positive period of the function  $y = \tan\left(\frac{1}{3}x + \frac{\pi}{4}\right)$ .)



## 自测题(整合习题)

### (Self-Test (Integrated Exercises))

#### 一、选择题(Multiple Choice)

1. 函数  $y = -\sin 6x$  ( $x \in \mathbb{R}$ ) 的最小正周期是 ( )。

(The minimum positive period of the function  $y = -\sin 6x$  ( $x \in \mathbb{R}$ ) is ( ).)

A.  $2\pi$  B.  $\pi$  C.  $\frac{\pi}{3}$  D.  $\frac{2\pi}{3}$

2.  $y = 3\cot \frac{x}{4}$  的最小正周期是 ( )。

(The minimum positive period of  $y = 3\cot \frac{x}{4}$  is ( ).)

A.  $4\pi$  B.  $3\pi$  C.  $\frac{1}{4}\pi$  D.  $2\pi$

3.  $y = 9\tan \frac{x}{3}$  的最小正周期是 ( )。

(The minimum positive period of  $y = 9\tan \frac{x}{3}$  is ( ).)

A.  $9\pi$  B.  $6\pi$  C.  $3\pi$  D.  $\pi$

4. 函数  $y = \cos x + 6$  是 ( )。

(The function  $y = \cos x + 6$  is a ( ) function.)

A. 奇函数(Odd) B. 偶函数(Even)

C. 既是奇函数又是偶函数(Both odd and even)

D. 非奇非偶函数(Neither odd nor even)

5. 函数  $y = \cos\left(x + \frac{\pi}{7}\right)$  是 ( )。

(The function  $y = \cos\left(x + \frac{\pi}{7}\right)$  is a ( ) function.)

A. 奇函数(Odd) B. 偶函数(Even)

C. 既是奇函数又是偶函数(Both odd and even)

D. 非奇非偶函数(Neither odd nor even)

6. 函数  $y = -3\sin x + 2$  是 ( )。

(The function  $y = -3\sin x + 2$  is a ( ) function.)

- A. 奇函数(Odd) B. 偶函数(Even)  
C. 既是奇函数又是偶函数(Both odd and even)  
D. 非奇非偶函数(Neither odd nor even)

7. 函数  $y = 4\sin\left(x + \frac{\pi}{6}\right)$  的奇偶性是 ( )。

(The parity of the function  $y = 4\sin\left(x + \frac{\pi}{6}\right)$  is ( ).)

- A. 奇函数(Odd) B. 偶函数(Even)  
C. 非奇非偶函数(Neither odd nor even)  
D. 减函数(Decreasing function)

8.  $y = 3\sin x$  的减区间是 ( )。

(The decreasing interval of  $y = 3\sin x$  is ( ).)

- A.  $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$  B.  $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$  C.  $[0, \pi]$  D.  $[\pi, 2\pi]$

9.  $\sin \frac{10\pi}{3} =$  ( )。

- A.  $\frac{1}{2}$  B.  $\frac{\sqrt{3}}{2}$  C.  $-\frac{1}{2}$  D.  $-\frac{\sqrt{3}}{2}$

## 二、填空题(Fill in the Blanks)

1.  $-130^\circ =$  \_\_\_\_\_  $\pi$ ;  $145^\circ =$  \_\_\_\_\_  $\pi$ ;

$\frac{11}{4}\pi =$  \_\_\_\_\_  $^\circ$ ;  $\frac{3\pi}{4} =$  \_\_\_\_\_  $^\circ$ 。

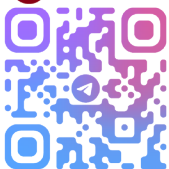
角  $1230^\circ$  在第\_\_\_\_\_象限。(The angle  $1230^\circ$  lies in the \_\_\_\_\_ quadrant.)

2. 若  $\alpha$  是钝角, 那么  $\frac{\alpha}{3}$  在第\_\_\_\_\_象限。

(If  $\alpha$  is an obtuse angle, then  $\frac{\alpha}{3}$  lies in the \_\_\_\_\_ quadrant.)

3.  $\sin 30^\circ =$  \_\_\_\_\_;  $\tan 30^\circ =$  \_\_\_\_\_;  $\tan 60^\circ =$  \_\_\_\_\_;  $\cos 420^\circ =$  \_\_\_\_\_。





4. 角  $\alpha$  的终边经过点  $(-2, -3)$ , 则  $\cot\alpha =$ \_\_\_\_\_。

(The terminal side of angle  $\alpha$  passes through point  $(-2, -3)$ , then  $\cot\alpha =$ .)

5. 求值:  $\sin\frac{8\pi}{6} =$ \_\_\_\_;  $\cos 600^\circ =$ \_\_\_\_;  $\tan(-225^\circ) =$ \_\_\_\_\_。

6.  $y = \frac{1}{3}\sin\left(3x + \frac{\pi}{4}\right)$  的最小正周期是\_\_\_\_\_;

最大值是\_\_\_\_;奇偶性是\_\_\_\_\_。

(The minimum positive period of  $y = \frac{1}{3}\sin\left(3x + \frac{\pi}{4}\right)$  is \_\_\_\_\_; its maximum value is \_\_\_\_\_; its parity is \_\_\_\_\_.)

7.  $y = 4\cos x$ ,  $x \in (-\pi, \pi)$  的减区间是\_\_\_\_\_。

(The decreasing interval of  $y = 4\cos x$  for  $x \in (-\pi, \pi)$  is \_\_\_\_\_.)

8.  $y = 4\tan\frac{1}{3}x$  的定义域是\_\_\_\_;最小正周期是\_\_\_\_\_。

(The domain of  $y = 4\tan\frac{1}{3}x$  is \_\_\_\_\_; its minimum positive period is \_\_\_\_\_.)

9.  $\tan 300^\circ + \cot 405^\circ =$ \_\_\_\_\_。

### 三、计算题(Calculation)

1. 已知  $\tan\alpha = 4$ , 求  $\frac{5\sin\alpha - 3\cos\alpha}{6\cos\alpha + 4\sin\alpha}$  的值。(Given  $\tan\alpha = 4$ , find the value of

$$\frac{5\sin\alpha - 3\cos\alpha}{6\cos\alpha + 4\sin\alpha}.)$$

2. 已知  $\cos\alpha = \frac{1}{5}$ ,  $\frac{3}{2}\pi < \alpha < 2\pi$ , 求  $\sin\alpha$  和  $\tan\alpha$  的值。(Given  $\cos\alpha = \frac{1}{5}$  and

$$\frac{3}{2}\pi < \alpha < 2\pi, \text{ find the values of } \sin\alpha \text{ and } \tan\alpha.)$$